

Abhirup Bhowmick

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PROFILE

Results-driven Software Engineer with a strong foundation in full-stack development, scalable backend systems, and modern AI integrations (RAG, Multimodal LLMs). Adept at bridging complex data structures with modern web technologies to deliver high-performance software solutions. Highly motivated problem-solver with a passion for clean system architecture, agile development, and continuous learning.

TECHNICAL ACHIEVEMENTS

- **TCS CodeVita Season 13:** Secured a **Global Rank of 2637**. Demonstrated elite proficiency in algorithms, data structures, and high-pressure analytical problem-solving.

TECHNICAL COMPETENCIES

- **Languages & Core CS:** Python, Java, JavaScript/TypeScript, SQL, C/C++, Algorithms, OOD, RESTful APIs.
- **Backend & Data:** Spring Boot, Node.js, PostgreSQL, MySQL, Neo4j (Knowledge Graphs).
- **Frontend & Cloud:** React.js, Next.js, Tailwind CSS, HTML5/CSS3, Docker, CI/CD, Edge Functions.
- **AI Integrations:** Vercel AI SDK, Multimodal LLMs (Gemini Pro), Retrieval-Augmented Generation (RAG).

PROFESSIONAL EXPERIENCE

IDEAS-TIH, Indian Statistical Institute (ISI)

Technical Engineering Associate

Kolkata, India

Aug 2025 – Sep 2025

- Engineered data analytics pipelines and developed high-performance backend services for enterprise AI/ML research applications.
- Translated complex, unstructured data into scalable technical designs and optimized SQL database schemas alongside senior researchers.
- Implemented robust testing and validation methodologies to ensure strict data integrity and system reliability in production environments.

Gameonix

6 Months

Software Development Intern

- Designed responsive UI components using modern frontend technologies and seamlessly integrated them with scalable backend services.
- Interpreted business requirements within Agile development cycles to deliver high-quality software features on tight deadlines.
- Troubleshoot application latency and resolved complex customer-facing software issues to enhance overall system reliability.

TECHNICAL PROJECTS

Visi Core AI — Full-Stack AI Platform

Live Demo

- Architected a serverless AI vision-understanding platform utilizing **Next.js**, **React**, and the **Vercel AI SDK**, enabling users to extract structured insights from complex visual data.
- Implemented real-time data streaming using Server-Sent Events (SSE) and Edge API routes, significantly reducing perceived latency during heavy **Multimodal LLM** processing.
- Engineered robust prompt pipelines enforcing strict JSON parsing, ensuring deterministic AI outputs and high fault tolerance for unpredictable edge cases.

ResuMatch AI — Automated SaaS Platform

Live Demo

- Engineered an AI-driven SaaS platform utilizing a secure **Java Spring Boot** and **PostgreSQL** backend, achieving 90%+ accuracy in automated resume-to-JD matching.
- Built and consumed secure RESTful APIs to handle complex business logic and manage a Razorpay webhook-integrated tiered subscription model.

Second Brain AI — Distributed Knowledge Graph

Live Demo

- Designed a cognitive platform using **React** and **TypeScript** that replaces flat folders with a **Neo4j Knowledge Graph**, enabling O(1) relationship traversal for interconnected data maps.
- Developed a Retrieval-Augmented Generation (RAG) pipeline utilizing **Dockerized microservices** to ground context-aware AI assistants securely in private user data.

EDUCATION

Brainware University — B.Tech in Computer Science Engineering (AI-ML)

2022 – 2026

- **GPA: 9.45 / 10.0** — Cumulative CGPA: 8.56 / 10.0.

CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) Generative AI Professional
- Oracle Cloud Infrastructure (OCI) AI Foundations
- Cisco CCNA: Introduction to Networks